Soumya Sharma

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Education

Birla Institute of Technology, Mesra (Patna Campus) — Patna, India
Bachelor of Technology in Electronics and Communication Engineering; CGPA: 7.2/10 — *2024 – May 2028*

Children College, Azamgarh — *Class XII (ISC): 93% | Class X (CISCE): 92.4%* — Azamgarh, India

Experience

NS Apps Innovation — Sep 2026 – Nov 2026
AI/ML & React Developer Intern — *India*

- Built the **ESP32-S3** (XIAO Sense) firmware for Yoda, a working AI desk robot: OV2640 hardware-JPEG frames double-buffered in **PSRAM** and posted zero-copy; a 16 kHz **I2S** PDM mic task pinned to core 0 (DC-blocking filter) feeds a 2 s **FreeRTOS** stream buffer so blocking uploads never drop samples
- Multiplexed audio and commands on one **WebSocket** to fit two TLS sessions (∼40–50 KB RAM each); added exponential backoff (0.5–8 s), an HTTP polling fallback and a firmware-side command allow-list
- Built the on-prem inference stack on an **NVIDIA DGX Spark** (ARM64), no cloud: **ONNX Runtime** pose, face-ID and emotion models (∼110 MB) on CPU at **60–70 ms/frame, 5 FPS, 0 dropped frames**; offline speech via Silero VAD, int8 faster-whisper (1.4 s per 2.5 s of speech) and Qwen3-VL. The robot greets people who wave, recognises enrolled teammates and answers "Hey Yoda" questions about what it sees
- Built a fully local **RAG** assistant (BGE-M3, Qdrant, Ollama) on a CPU-only machine, benchmarking gemma3:4b (∼10 tok/s) vs. llama3.2:1b (∼21 tok/s); a knowledge-graph tier answers in **0.02–0.13 s** with no LLM

Undergraduate Research Intern – Nanopore Biosensor Signal Processing — Aug 2026 – Present
Under Dr. Naren Das, Dept. of ECE, BIT Mesra — *Patna, India*

- Built a **signal-processing** pipeline for nanopore current traces: rolling-median baseline, noise estimation, threshold + hysteresis event detection, AdaBoost event fitting and **PSD noise spectroscopy** + PLS
- Validated on 3,150 synthetic events (**92–100%** shape accuracy at ≥4× rise time) and 440 real recordings (3.1 GB); event classifier reached **0.84 balanced accuracy** on an independent day vs. 0.55 baseline

Projects

ADAS for an Autonomous Mini Vehicle | *YOLOv8n, TensorFlow Lite, Raspberry Pi, ESP32, OpenCV* GitHub

- Deployed **YOLOv8n** via **TensorFlow Lite** (uint8 input) on a Raspberry Pi for real-time obstacle detection, using frame skipping and reduced 160/256 px inputs to keep inference on-board
- Linked on-board perception to an **ESP32** motor controller over **UART** (115200 baud) driving DC motors through an H-bridge with detection-driven drive commands

ML-Based MPPT Controller for a Grid-Tied Solar Inverter | *Python, NumPy, SciPy, MATLAB* — 2026

- Combined Gaussian-process Bayesian optimization with a **9-24-16-5 MLP (∼770 weights)** sized for an embedded control loop; the MLP flags when a global-peak search is needed (**98.6%** test accuracy, 31k samples)
- In digital-twin simulation, reached **99.37% mean tracking efficiency** across 10 irradiance, temperature and partial-shading scenarios vs. 96.89% for the existing P&O controller

Technical Skills

Languages: C/C++, Python, MATLAB, TypeScript
Embedded: ESP32-S3 (Arduino-ESP32), FreeRTOS, I2S, UART, PSRAM, WebSockets, TLS, Raspberry Pi
Edge AI: TensorFlow Lite, ONNX Runtime, YOLOv8, OpenCV DNN, int8 faster-whisper, Silero VAD
ML / Signal Processing: PyTorch, scikit-learn, LightGBM, NumPy, SciPy, PSD / noise analysis, Ollama

Applied Patent and Publications

- **Hybrid Psychometric-Machine Learning Platform for Multi-Modal Mental Health Screening of Engineering Students**: IRT-based adaptive mental-health screening

Achievements & Leadership

- **Finalist**, TantraFiesta 2025 (IIIT Nagpur) and Coinbase-sponsored Web3 Hackathon (BIT Mesra); Round 1, **Adobe Hackathon 2026**
- **Mentoring**: conducted ML workshops (tutored ∼300 students) and a hands-on Arduino workshop (∼150 students)
- **Head of Publicity, TECHNIKA 26-27** (∼2,000 participants); **Ignite** Designing & Drafting Team
- **Basketball** gold, 2× inter-branch and inter-college; **Chess** gold, inter-branch; guitarist, Pratibimb Society (**1st prize**, band competition)